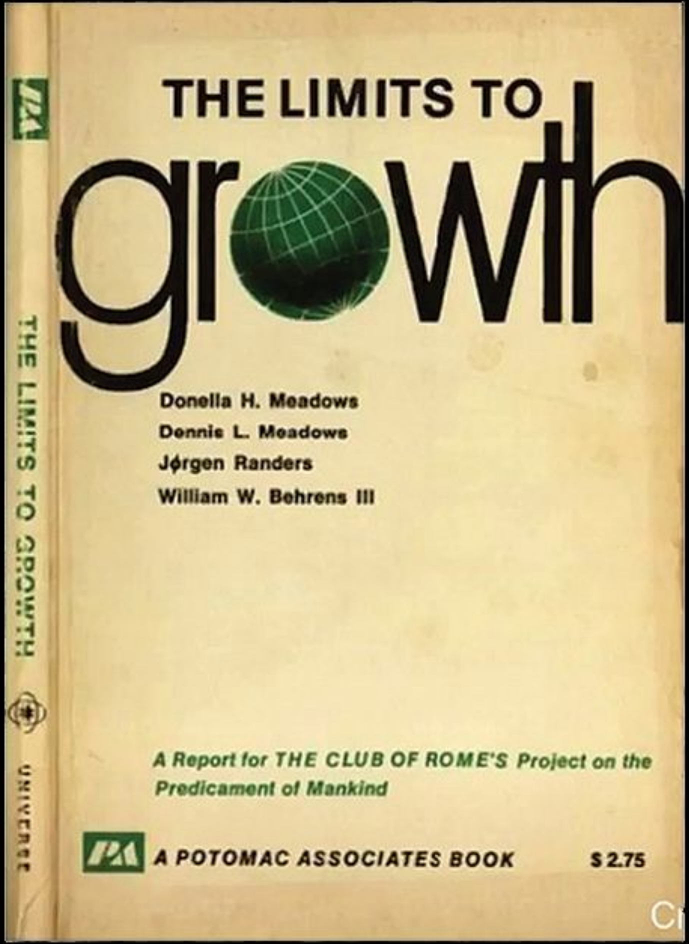
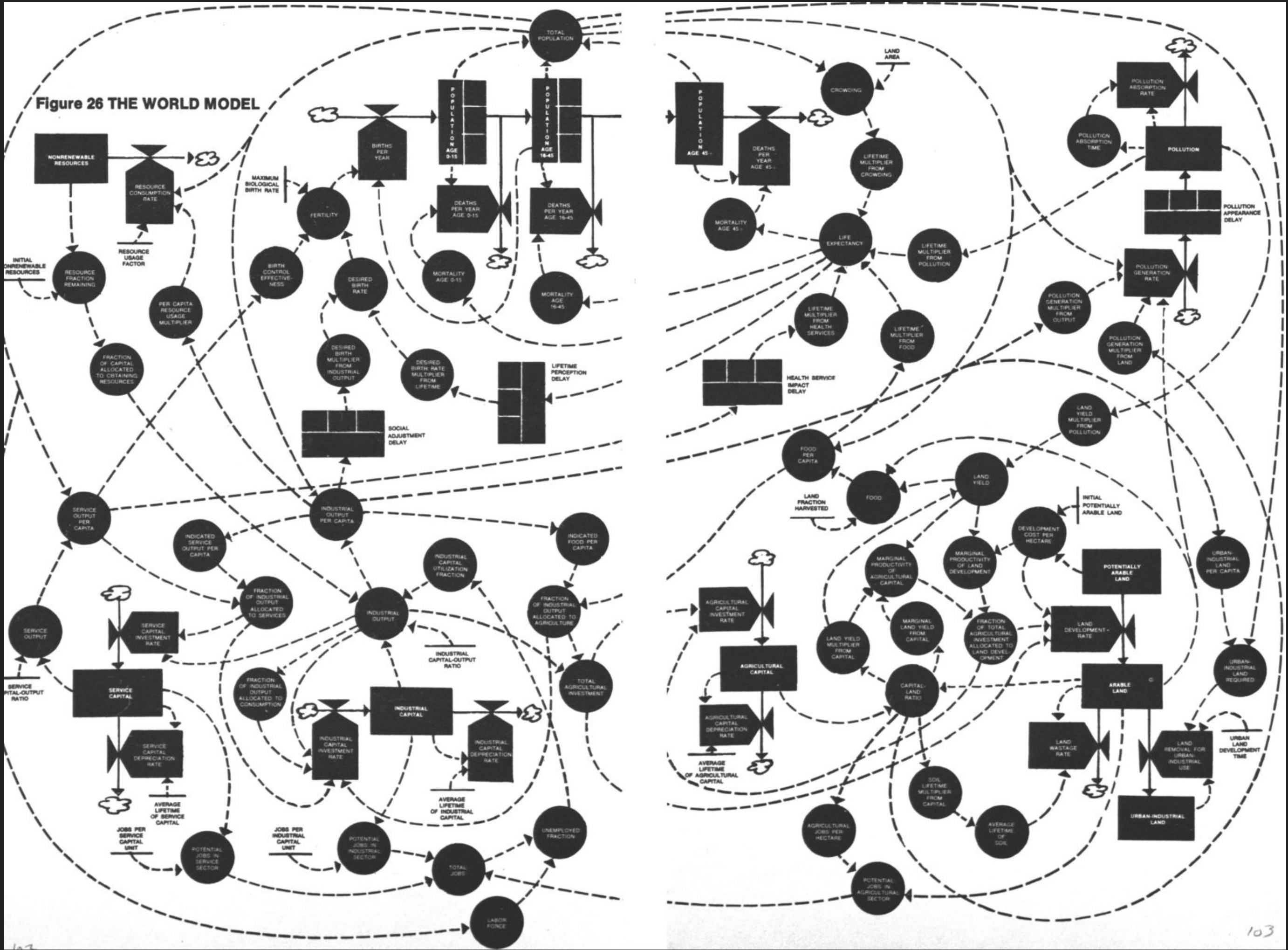
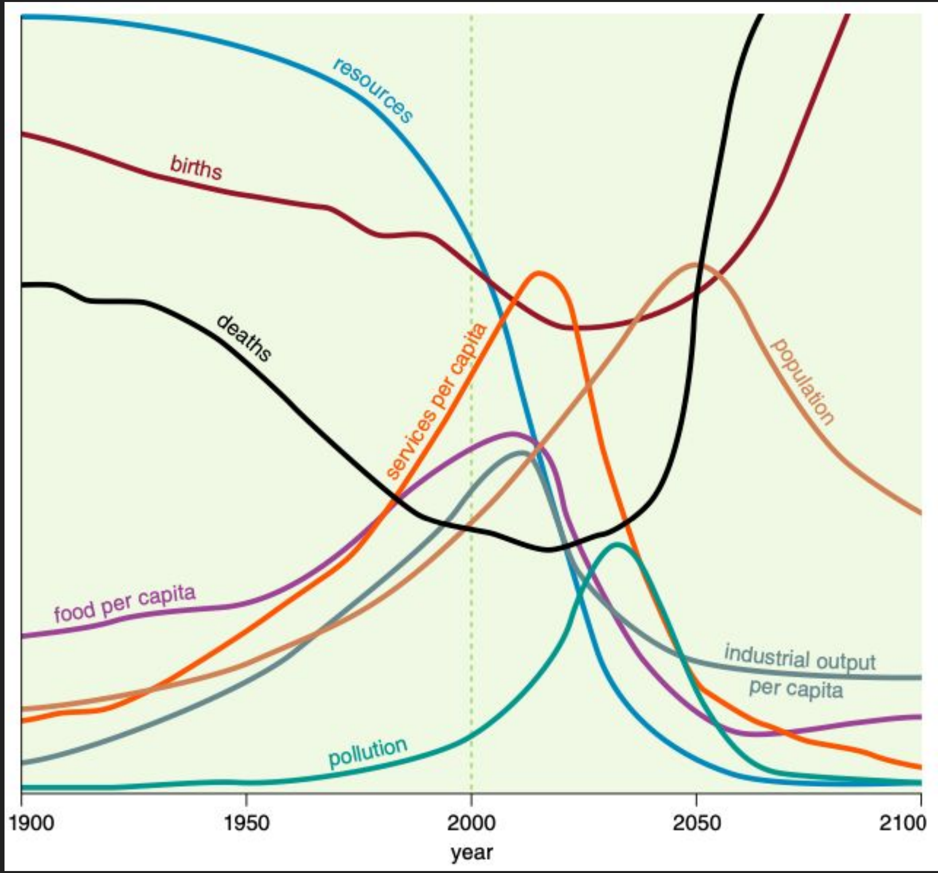
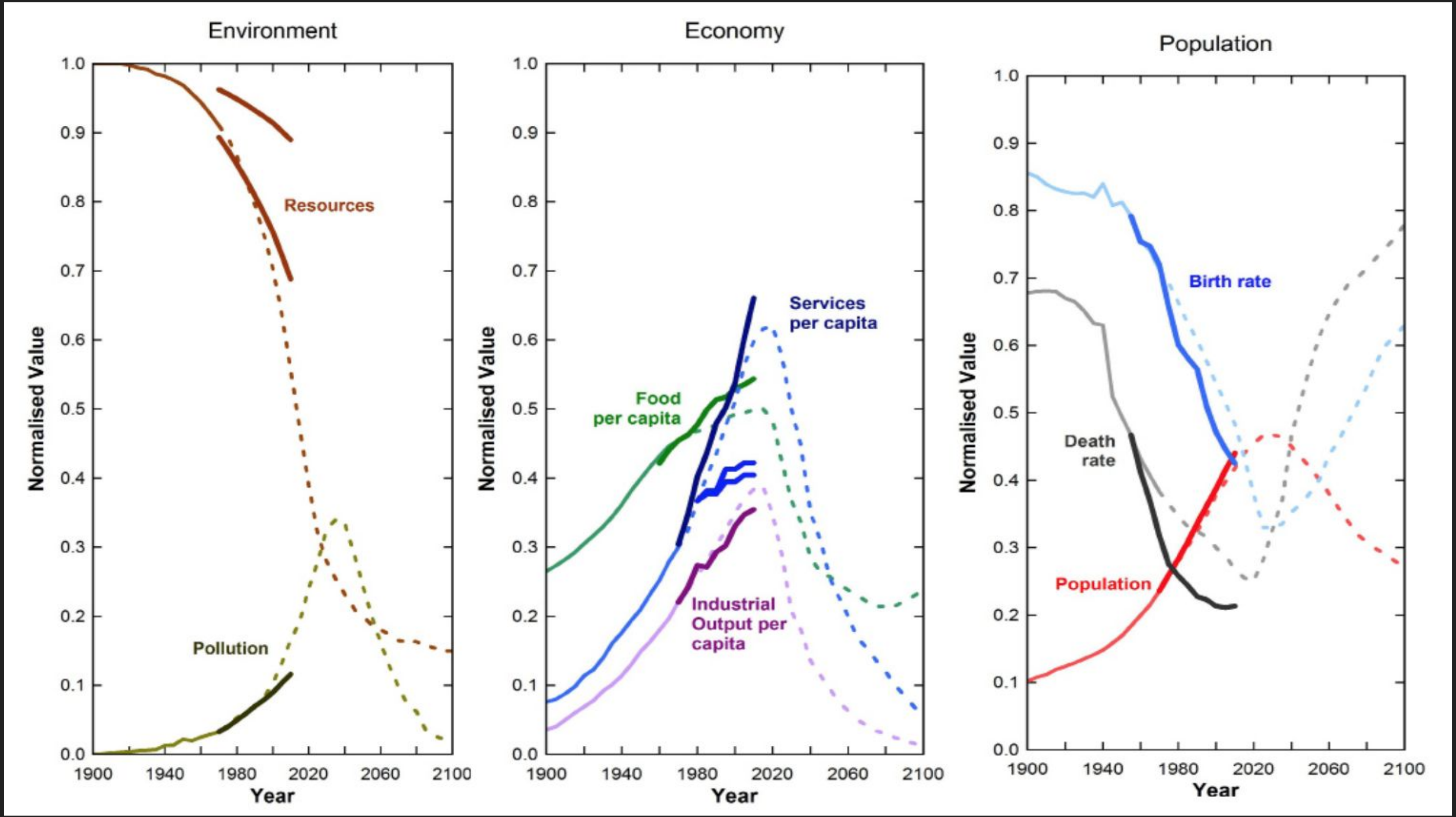

THE LIMITS TO GROWTH
UNIVERSE
THE LIMITS TO
growth
Donella H. Meadows
Dennis L. Meadows
Jørgen Randers
William W. Behrens III
A Report for THE CLUB OF ROME'S Project on the Predicament of Mankind
A POTOMAC ASSOCIATES BOOK
$ 2.75

Figure 26 THE WORLD MODEL

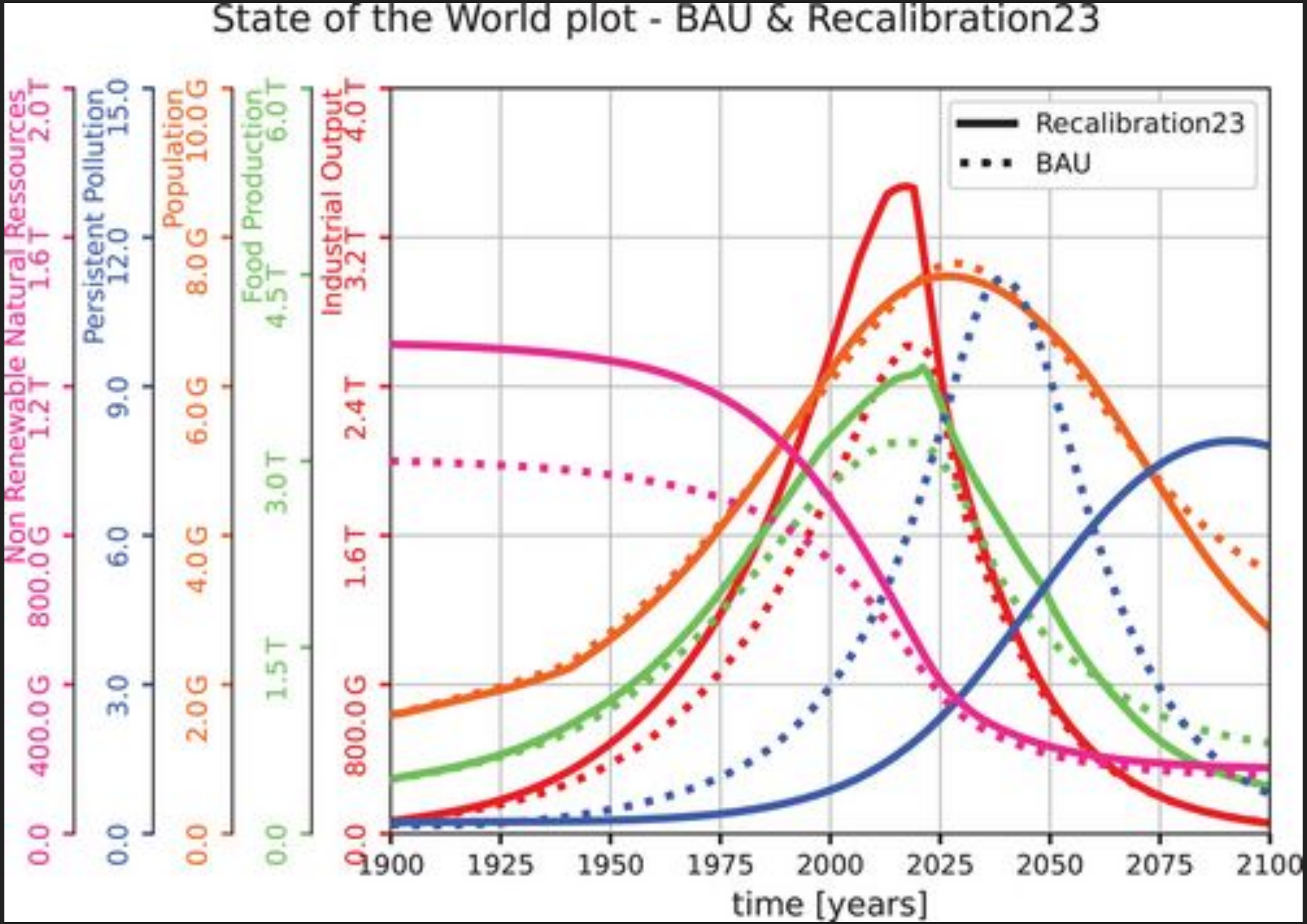


Overshoot & collapse

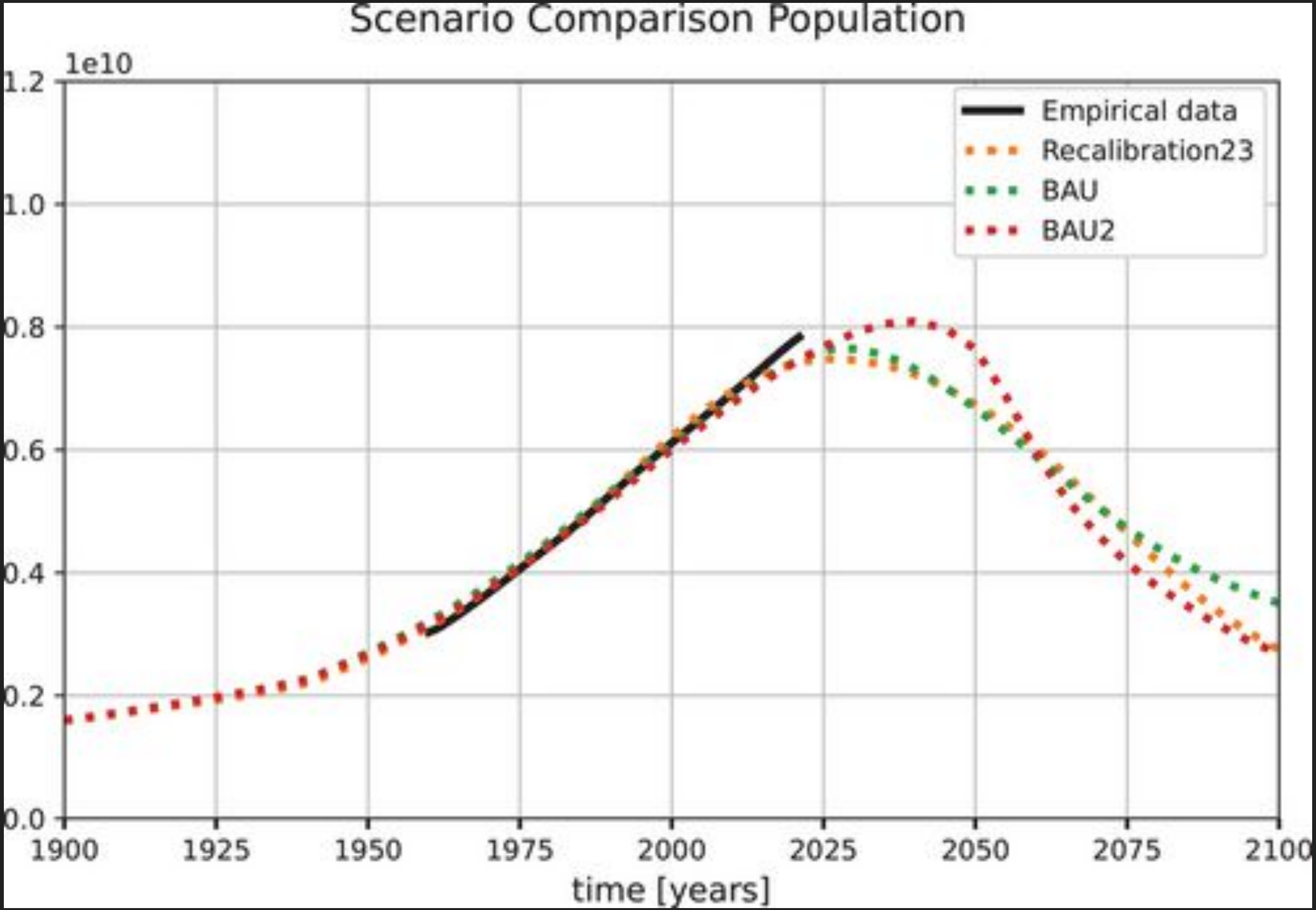




https://www.researchgate.net/publication/267751719_Is_Global_Collapse_Imminent_An_Updated_Comparison_of_The_Limits_to_Growth_with_Historical_Data



https://onlinelibrary.wiley.com/doi/10.1111/jiec.13442



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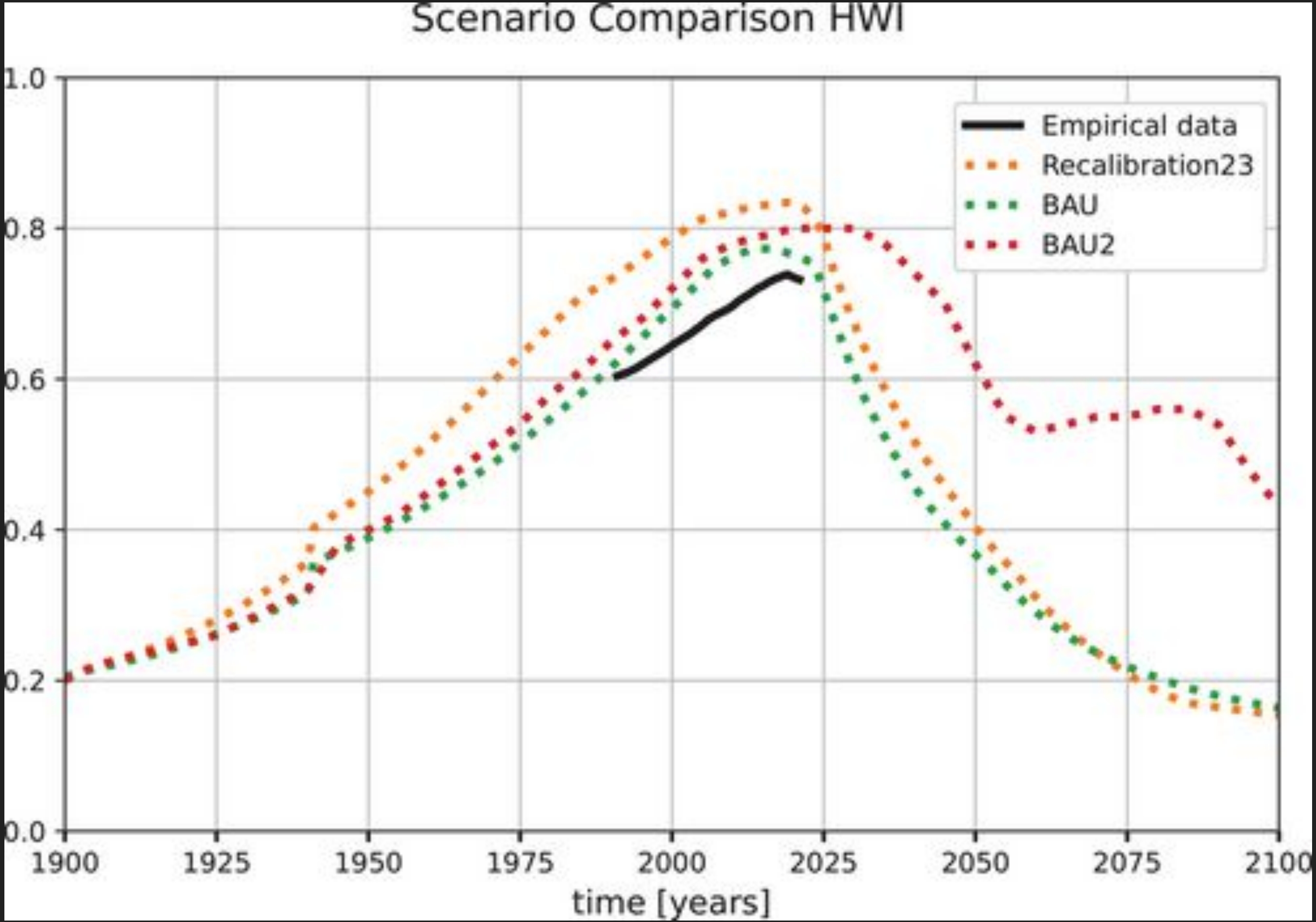


https://onlinelibrary.wiley.com/doi/10.1111/jiec.13442

IPCC on overshoot

- 3.3–3.6b people live in contexts that are highly vulnerable to climate change
- For 127 identified key risks, assessed mid- and long- term impacts are up to multiple times higher than currently observed
- There is increased evidence of maladaptation across many sectors and regions since the AR5
- Action is more urgent than previously assessed in AR5
- Climate-related migrants are not officially recognised as refugees under the 1951 Convention relating to the Status of Refugees

https://www.ipcc.ch/report/ar6/wg2/downloads/report/IPCC_AR6_WGII_FullReport.pdf

IPCC on overshoot

Planet's first catastrophic climate tipping point reached, report says, with coral reefs facing 'widespread dieback'

Unless global heating is reduced to 1.2C 'as fast as possible', warm water coral reefs will not remain 'at any meaningful scale', a report by 160 scientists from 23 countries warns

https://www.theguardian.com/environment/2025/oct/13/coral-reefs-ice-sheets-amazon-rainforest-tipping-point-global-heating-scientists-report

AMOC collapse

Nature. 2025 Feb 26;638(8052):987–994. doi: 10.1038/s41586-024-08544-0

Climate models suggest that the Atlantic Meridional Overturning Circulation is unlikely to collapse this century, owing to stabilization from wind-driven upwelling in the Southern Ocean.

https://pmc.ncbi.nlm.nih.gov/articles/PMC11864975/

AMOC collapse

nature > communications earth & environment > articles > article

Article | Open access | Published: 27 March 2026

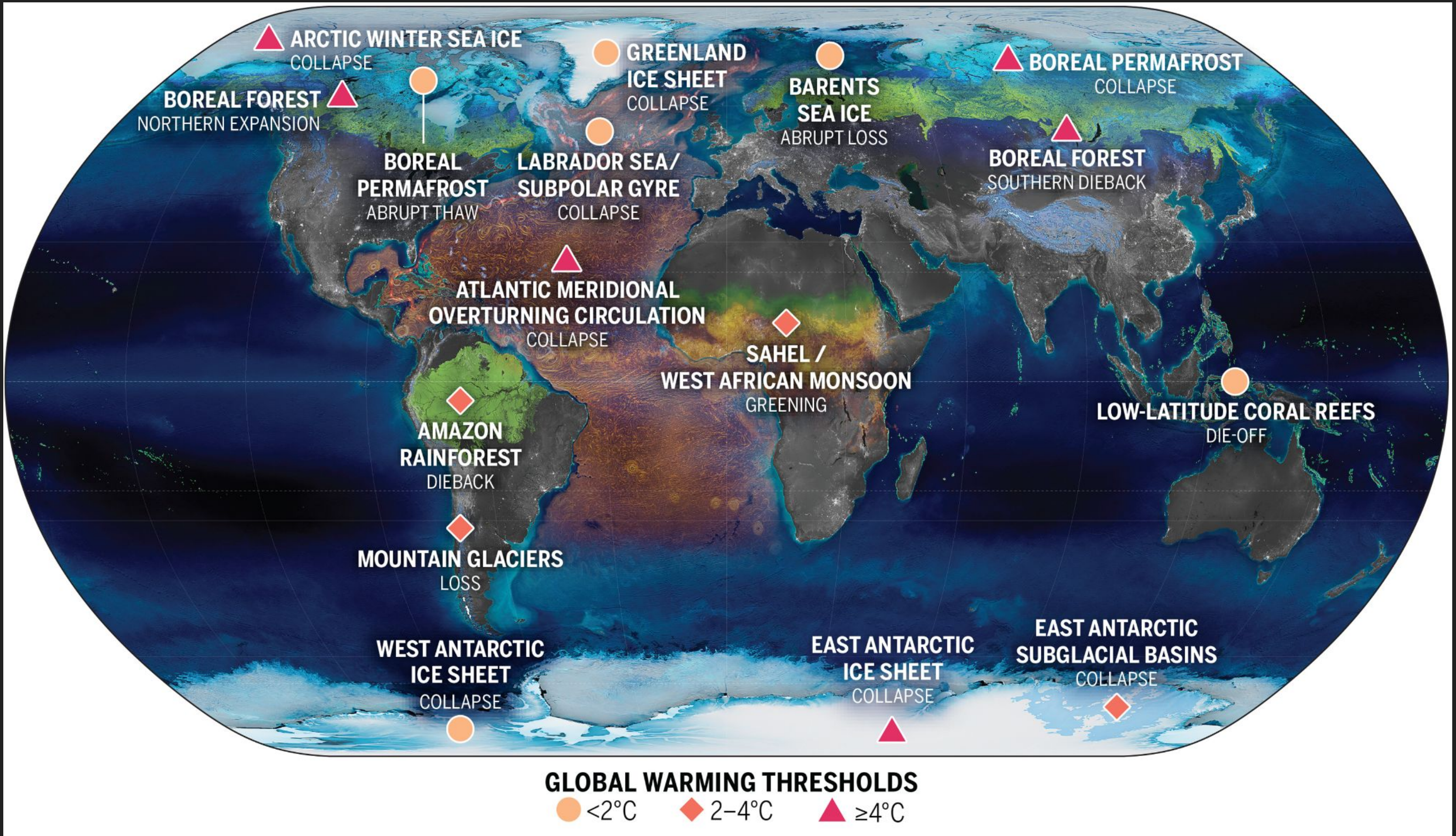
of such reconstructions is still debated[31]. The IPCC AR6 concludes that, with medium confidence, AMOC shutdown is unlikely this century[32]. However, recent research suggests this risk may be underestimated, highlighting the potential for substantial, persistent global impacts if tipping points were crossed[20,33,34]. Modeling studies also show that the weakening or shutdown of the AMOC can affect the functioning of the marine carbon cycle and cause additional changes in CO_2[35,36,37].

https://www.nature.com/articles/s43247-026-03427-w

Thwaites glacier collapse

"We really, really need to understand how fast the ice is changing, how fast it is going to change over the next 20 to 50 years," said Christine Dow, an associate professor of glaciology at the University of Waterloo and one of the study's authors. "We were hoping it would take a hundred, 500 years to lose that ice. A big concern right now is if it happens much faster than that."

https://www.pnas.org/doi/full/10.1073/pnas.2404766121



https://www.science.org/doi/10.1126/science.abn7950

CONCLUSION

Our assessment provides strong scientific evidence for urgent action to mitigate climate change. We show that even the Paris Agreement goal of limiting warming to well below 2°C and preferably 1.5°C is not safe as 1.5°C and above risks crossing multiple tipping points. Crossing these CTPs can generate positive feedbacks that increase the likelihood of crossing other CTPs. Currently the world is heading toward ~2 to 3°C of global warming; at best, if all net-zero pledges and nationally determined contributions are implemented it could reach just below 2°C. This would lower tipping point risks somewhat but would still be dangerous as it could trigger multiple climate tipping points.

https://www.science.org/doi/10.1126/science.abn7950

- Climate change feedbacks
- Cascade effects

reaching effects, potentially affecting political and social stability worldwide[2,3,4,5,6]. A notable example illustrating the complex interplay between climate change and socio-economic stability is the Arab Spring and its cascading impacts on the European Union (EU). Extreme weather caused crop failure at many locations globally in 2010 and 2011[7]. This shortfall in staple food production was further exacerbated by an export ban of grains by Russia, the high demand for biofuel crops and copycat investor behaviour in the financial commodity markets ('herding'), which drove up crop prices, ultimately leading to a food crisis[8,9]. Countries heavily reliant on crop imports, already grappling with poverty, inequality, weak governance and a history of conflicts, found themselves particularly vulnerable to higher food prices[7,10]. The additional societal pressure, in the context of historical state repression, contributed to the outbreak of conflicts and civil war in the Middle East and North Africa, which in the EU resulted in an increase in refugees and associated political fallout regarding immigration[11,12].

https://www.nature.com/articles/s41558-025-02455-2

The post-sustainability trilemma

1. economic growth
2. participation (democracy)
3. environmental protection

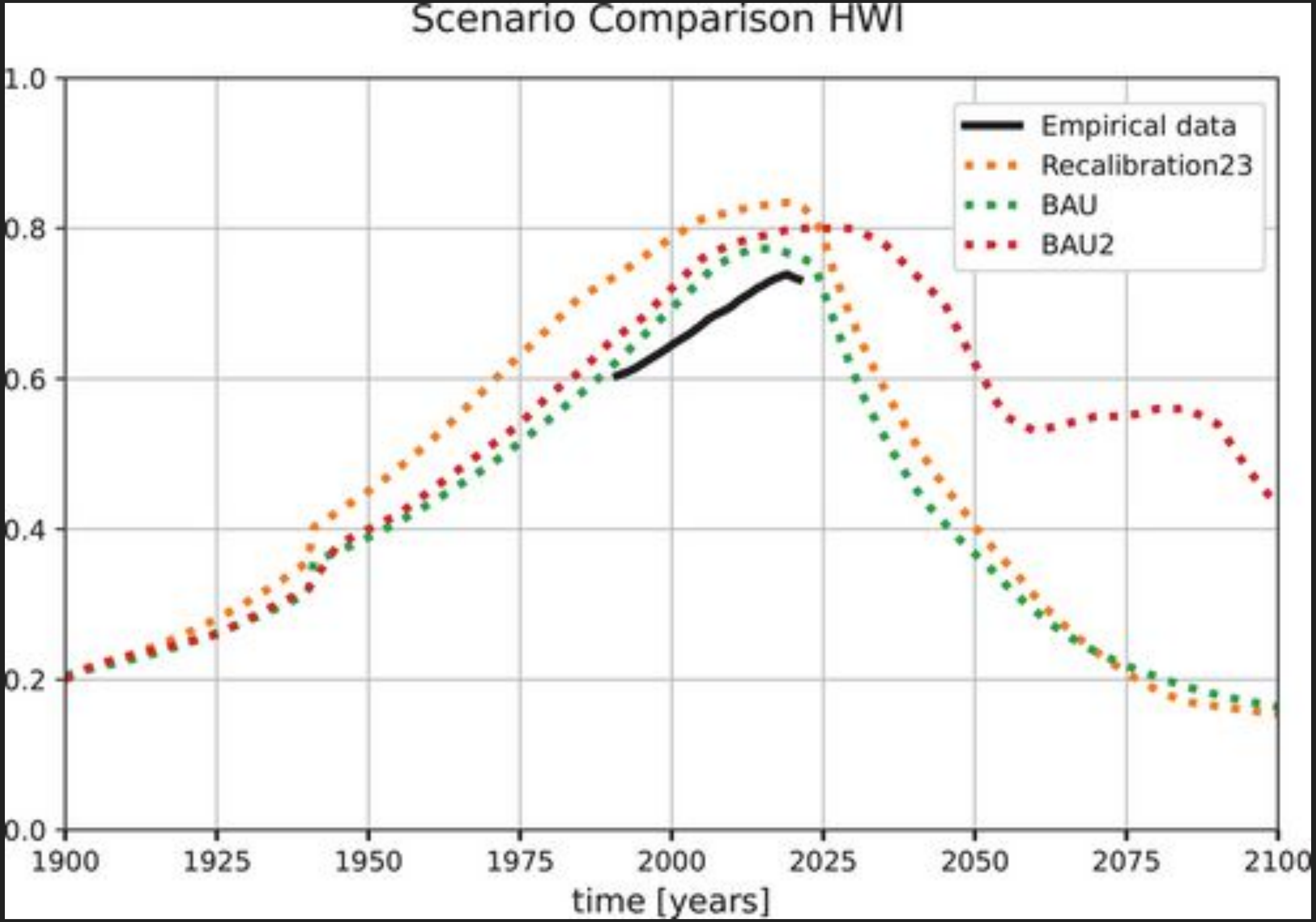
https://www.tandfonline.com/doi/full/10.1080/1523908X.2019.1673156

The post-sustainability trilemma

1. economic growth
2. participation (democracy)
3. environmental protection

- (1)–(2) techno business-as-usual
- (2)–(3) post-growth approaches
- (1)–(3) environmental authoritarianism

https://www.tandfonline.com/doi/full/10.1080/1523908X.2019.1673156



https://onlinelibrary.wiley.com/doi/10.1111/jiec.13442

Degrowth

As the climate crisis worsens and the carbon budgets set out by the Paris Agreement shrink, climate scientists and ecologists have increasingly come to highlight economic growth as a matter of concern. Growth drives energy demand up and makes it significantly more difficult – and likely infeasible – for nations to transition to clean energy quickly enough to prevent potentially catastrophic levels of global warming. In recent years, IPCC scientists have argued that the only feasible way to meet the Paris Agreement targets is to actively scale down the material throughput of the global economy. Reducing material throughput reduces energy demand, which makes it easier to accomplish the transition to clean energy.

https://www.paecon.net/PAEReview/issue87/Hickel87.pdf

Degrowth

- Reduced working hours
- Caps on resource use and emissions
- Universal Basic Income / Basic Services
- Maximum income caps
- Replace GDP with wellbeing indicators
- Not-for-profit cooperatives
- Public commons
- Ecovillages and Transition Towns
- Carbon taxes with dividends
- Caps on advertising

Permaculture

Permaculture is a design system modeled on natural ecosystems — productive, resilient, low-input.
Developed by Bill Mollison and David Holmgren in Australia in the 1970s
Core ethics: Earth Care, People Care, Fair Share

- Catch and store energy and water locally
- Use and value diversity
- Use small and slow solutions
- Integrate rather than segregate

https://en.wikipedia.org/wiki/Permaculture

Permaculture

- Food forests and no-till growing — sequester carbon and build soil
- Rainwater harvesting and greywater recycling
- Community-scale food production reducing supply chain dependence
- Regenerative agriculture rebuilding degraded land
- Rewilding margins and corridors to restore ecological function

https://en.wikipedia.org/wiki/Permaculture

Deep Adaptation

- Near-term societal collapse due to climate change is now likely, if not inevitable
- Aimed originally at sustainability professionals
- Denial (in science, institutions, and individuals) has kept this conversation suppressed
- Hopelessness and grief, if properly held, can be transformative

https://www.lifeworth.com/deepadaptation.pdf

The Four Rs

- Resilience: What do we want to keep?
  - Preserve valued norms, relationships, and ways of life through disruption
  - Psychological resilience: bouncing back does not mean returning to "normal"
- Relinquishment: What do we need to let go of?
  - Coastlines, high-risk industries, unsustainable consumption expectations
  - Also letting go of the growth paradigm itself

https://www.lifeworth.com/deepadaptation.pdf

The Four Rs

- Restoration: What can we bring back?
  - Seasonal diets, local production, community mutual aid, rewilded land
  - The practices of permaculture and degrowth belong here
- Reconciliation: With whom and what can we make peace?
  - With each other, with loss, with mortality, avoiding harm from suppressed panic
  - Reconciling with the limits that Meadows, Randers, and the Club of Rome named in 1972

All things at once

- *Limits to Growth:* exponential growth on a finite planet leads to overshoot and collapse, and the data now confirms we are on that trajectory
- *Tipping points:* some of those limits have already been crossed; we are now inside the feedback loops
- *IPCC projections:* the physical consequences are locked in across multiple timescales, and every fraction of a degree matters
- *Climate action:* positive change is accelerating, but not yet fast enough
- *Permaculture:* practical tools for living and producing within ecological limits
- *Degrowth:* a political and economic framework for a planned, equitable transition beyond the growth paradigm
- *Sustainability critique:* can growth ever be truly decoupled from ecological harm?
- *Deep Adaptation:* the honest reckoning when everything else may be too late, and a map for navigating tragedy with integrity

All things at once

- **Understand the full picture.** From Limits to Growth to tipping points to Deep Adaptation
- **Grieve honestly.** Suppressing awareness makes collective action harder
- **Build local resilience.** Food, community, skills, relationships, permaculture
- **Practice relinquishment.** Let go of growth as the organising principle of life
- **Support restoration.** Rewilding, regenerative agriculture, local economies and commons
- **Advocate for degrowth policies.** Shorter work weeks, wellbeing indicators, universal basic services
- **Seek reconciliation.** With others, with nature, with uncertainty and loss
- **Stay in the conversation.** The worst outcome is silence and isolation

https://github.com/hom3chuk/ltg-talk